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SARDAR PATEL UNIVERSITY
T.Y.B.C.A. 6th SEMESTER CBCS Examination-2024
US06BCA53 || Computer Networks



Date: 30/03/2024 Saturday

Time: 10:00am to 01:00pm

Maximum Marks: 70

Q.1

Answer the following questions.

[10]

- 1 When the packets are small and all the same size, they are often called _____.
(a) Cells (b) Trunks (c) Circuits (d) Subnet
- 2 Traditional LANs run at speeds of _____ to _____ Mbps.
(a) 20 to 100 (b) 10 to 100 (c) 10 to 1000 (d) 100 to 1000
- 3 "Interconnected collection of autonomous computers" is known as _____.
(a) Host (b) Subnet (c) Computer network (d) Server
- 4 A _____ cable is made of glass or plastic and transmits signals in the form of light.
(a) Fiber Optical (b) Coaxial cables (c) Twisted-pair (d) None of these.
- 5 _____ is an analog multiplexing technique that combines analog signals.
(a) TDM (b) FDM (c) WDM (d) None of these.
- 6 In a fiber-optic cable, the signal is propagated along the inner core by _____.
(a) Refraction (b) Modulation (c) Reflection (d) Amplification
- 7 A _____ is a set of rules that govern data communications.
(a) Layers (b) Network (c) Communication (d) Protocol
- 8 The signal from a satellite is normally aimed at a specific area called _____.
(a) Channel (b) Tower (c) Orbit (d) Footprint
- 9 Which of the following topology requires maximum cabling?
(a) Mesh (b) Bus (c) Star (d) Ring
- 10 A switch performs filtering on the basis of _____ address.
(a) MAC (b) logical (c) port (d) postal

Q.2

Explain following in brief. (Attempt any Ten)

[20]

- 1 What is Bandwidth?
- 2 What are the uses of computer networks?
- 3 Explain SUBNET in brief.
- 4 Explain Magnetic media in short.
- 5 Explain application of twisted pair in short.
- 6 Explain serial transmission in short.
- 7 Explain orbit.
- 8 Define: Protocol Stack
- 9 What is satellite? List categories of satellite.
- 10 Explain Bluetooth in short.
- 11 Explain Wireless Network in short.
- 12 Explain Gateway in short.

(1)

[P.T.O.]

Q-3 What are the advantages & Disadvantages of computer networks? Explain in detail. **[10]**

OR

Q-3 How can we classify computer Network? Explain in detail. **[10]**

Q-4 (A) Explain FDM in detail. **[05]**

(B) Explain coaxial cable in detail with its performance and applications. **[05]**

OR

Q-4 (A) Explain Synchronous TDM in detail. **[05]**

(B) Explain fiber optic in detail. **[05]**

Q-5 (A) Write a note on protocol hierarchies. **[05]**

(B) List all layers of OSI model. Explain any two layers. **[05]**

OR

Q-5 (A) Write a note on design issues for the layers. **[05]**

(B) Write a note on satellite Network. **[05]**

Q-6 (A) Explain Bus & Ring topology in detail. **[05]**

(B) Explain Modem & Repeater & Amplifier in detail. **[05]**

OR

Q-6 (A) Write a short note on CSMA. **[05]**

(B) Explain Mesh & Star topology in detail. **[05]**

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